

Muhammad Abdullah Goher

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Education

University of Pennsylvania

Aug 2023 – May 2027

B.S.E. & M.S.E. in Computer Science (Submatriculation), Concentration in Artificial Intelligence

Philadelphia, PA

GPA: 3.72 (B.S.E.) / 3.88 (M.S.E.) | **Coursework:** Applied Machine Learning, Artificial Intelligence, Big Data Analytics, Computer Graphics, Operating Systems, Databases, Probability, Statistical Inference

Experience

AeternalLabs, PBC – Founding AI Engineer

May 2026 – Present

Runtime AI-governance platform auditing LLMs for demographic fairness

Saint Louis, MO

- Built a low-latency bias detector screening LLM outputs across **10+ regulated domains** to catch discriminatory automated decisions in production.
- Integrated a multi-provider LLM routing layer spanning **8 providers and 18+ frontier models** (OpenAI, Anthropic, Google, xAI, DeepSeek, Cerebras) behind a single FastAPI gateway.
- Parallelized counterfactual model evaluation with a thread pool, cutting batch runtime $\sim 2\times$ (10+ hours saved on large runs).
- Designed a 3NF PostgreSQL audit trail (SQLAlchemy) with SHA-256 integrity hashing and JWT + Google OAuth for tamper-evident, access-controlled records.

Computational Social Science Lab (Prof. Duncan Watts) – Data Science Research Assistant May 2025 – Dec 2025

Human-mobility research platform & large-scale geospatial analysis

Philadelphia, PA

- Built an LLM-powered pipeline extracting structured JSON features from **100+** scholarly papers, plus a RAG layer surfacing quantitative mobility metrics.
- Modeled crowd density and flow from **multi-terabyte** GPS datasets (Meta, Factori, SafeGraph) with PySpark on AWS EMR, supporting Disaster Management planning for the 2026 FIFA World Cup.

PennAdapt – Machine Learning Engineer

Sep 2024 – Present

Penn Assistive Devices & Prosthetic Technologies

Philadelphia, PA

- Built a real-time computer-vision model (CNN + Vision Transformer, PyTorch) detecting **50+ classes** of surgical tools at **90%+** accuracy across varying angles.
- Designed a multi-layer agentic pipeline chaining SOTA models (SAM segmentation + ViT classification) with a RAG parent LLM that verifies and approves the final tool count.
- Curated and annotated a **1,500+** image dataset with Penn Medicine surgeons within a 10-engineer team.

Clab AI – Artificial Intelligence Intern

May 2024 – Aug 2024

AI-powered college-application assistant used by 100+ students

Hybrid – Nashville, TN

- Fine-tuned LLMs on accepted essays with RLHF, and built hyperparameter-tuned Random Forest / Linear Regression models predicting financial-aid eligibility at **82% R^2** (EDA across 100+ universities).

Projects

Research Swarm | Python, LangGraph, LangChain, FastAPI

- Built a LangGraph multi-agent research system (converge-diverge): a planner dynamically fans out parallel research branches, then a *cross-vendor* critic verifies key claims to cut correlated hallucination.
- Orchestrated 5+ LLM providers (Claude, GPT o-series, Gemini, Perplexity, local Ollama) by task strength, with budget- and round-capped re-diverge loops feeding a cited synthesis.

News Source Classification | PyTorch, HuggingFace Transformers, scikit-learn

- Co-built (team of 3) a headline classifier distinguishing Fox vs. NBC; scaled the dataset **3,800** $\rightarrow$ **17,900** headlines via multithreaded sitemap and Apify scraping pipelines.
- Fine-tuned RoBERTa/DeBERTa backbones; best model (RoBERTa-base + LinearSVC head on [CLS] embeddings) reached **91.75% accuracy**, +25 pts over a TF-IDF baseline.

Technical Skills

AI / ML: PyTorch, TensorFlow/Keras, scikit-learn, HuggingFace Transformers, LLM fine-tuning (LoRA/PEFT, RLHF), RAG, LangChain, LangGraph, agentic systems, CNNs/ViTs, NLP, XGBoost

Languages: Python, C, C++, Java, JavaScript/TypeScript, SQL

Data: pandas, NumPy, PySpark, vector databases (FAISS/Pinecone/Chroma), Matplotlib, Jupyter

Backend & Infra: FastAPI, Node.js, Express, PostgreSQL, MongoDB, Docker, AWS (S3, EMR), Git, Linux